

Explainable AI for Exclusive Mother's Milk Feeding at Discharge in Baby-Friendly NICU

Elif Keles^{1,2}, Umut Cagin Ozcan¹, İbrahim Murat Hirfanoğlu², Esra Önal², Ebru Ergenekon², Esin Koç², Ulas Bagci^{1,3,4}

¹ Northwestern University, Feinberg School of Medicine, Department of Radiology, Chicago, IL¹

² Division of Neonatology, Gazi University, Faculty of Medicine, Ankara, Turkey

³ Northwestern University, Department of Biomedical Engineering , and ⁴Department of Electrical and Computer Engineering

Introduction

Human milk feeding is a core quality target in Baby-Friendly NICUs (BF-NICUs) and is associated with lower necrotizing enterocolitis, fewer infections, and improved neurodevelopment in medically fragile infants. Implementation intensity can vary across epochs (pre-certification, post-certification, COVID-19 pandemic). In parallel, modern NICUs increasingly deploy AI-enabled decision support for risk stratification, early-warning surveillance, imaging triage, and workflow orchestration in high-acuity settings. To enable precision, epoch-aware implementation in a Level IV NICU, we developed models to predict discharge feeding type and identify infants at risk for non-exclusive human milk feeding.

Methods

This was an IRB-approved study of 1,064 consecutive Level IV NICU admissions (length of stay ≥ 3 days). We prospectively collected 87 maternal, neonatal, and hospitalization variables across three predefined epochs (pre-BF-NICU certification, post-certification/pre-pandemic, pandemic). Outcomes were exclusive human milk, mixed human milk plus formula, or other. Using stratified sampling, data were split into training (80%) and test (20%) sets. Random Forest, XGBoost, and multinomial logistic regression were compared with 5-fold stratified cross-validation and nested hyperparameter tuning. Test performance was summarized with accuracy, weighted-F1, and AUROC. Explainability was performed using SHAP.

Results

Infants spanned 22–42 weeks' gestation with NICU stays of 3–299 days. On the independent test set, Random Forest performed best (accuracy **0.84**, weighted-F1 **0.83**, AUROC **0.90**), followed by XGBoost (**0.81/0.81/0.88**) and multinomial logistic regression (**0.75/0.76/0.85**). Discrimination between exclusive and mixed feeding was robust and consistent across epochs, while predictions for the minority “other” class were limited by class imbalance. Global SHAP analysis indicated that early in-hospital feeding patterns were most influential—specifically, day-1 to day-3 volumes of formula and mother's own milk; the route of feeding on day 3 (oral vs gavage); total enteral intake on day 3; the indicator for combined feeding route (oral plus gavage); maternal age; and birthweight. Class-specific explanations showed that predictions for exclusive human milk were primarily shaped by higher early exposure to mother's milk and documented breastfeeding education status; mixed feeding predictions were driven by early formula exposure, the day-3 feeding route, and birthweight; and other feeding predictions were influenced by mother's milk exposure on day 3, the day-3 feeding route, day-2 feeding volumes,

and maternal age. These SHAP summaries reflect model contribution magnitudes only and do not imply clinical causality.

Discussion

Tree-based ensembles yielded reliable, interpretable predictions of discharge feeding type in a high-risk Level IV NICU. Embedding the risk score within the electronic health record as an AI-enabled, human-in-the-loop tool can (i) stimulate earlier bedside breastfeeding plans, (ii) standardize donor-milk counseling and skin-to-skin facilitation despite isolation constraints, (iii) prioritize high-risk dyads during staffing shortages, and (iv) integrate with other NICU AI systems to streamline workflows.

Conclusion

Random-Forest-based prediction (accuracy 0.84; weighted-F1 0.83; AUROC 0.90), coupled with SHAP explainability and EHR integration, can operationalize BF-NICU priorities by identifying infants at risk for non-exclusive human milk feeding and enabling timely interventions. Prospective implementation and strategies to address class imbalance are warranted to maximize exclusive human milk feeding and improve downstream neonatal outcomes.